

Putnam POTD November 16, 2025

I. PROBLEM

(1977 B5) Prove that for $n \geq 2$,

$$\underbrace{2^{2^{\dots^2}}}_{n \text{ terms}} \equiv \underbrace{2^{2^{\dots^2}}}_{n-1 \text{ terms}} \pmod{n}.$$

II. SOLUTION

We redefine the variables for style. Let

$$x_i = \underbrace{2^{2^{\dots^2}}}_{i \text{ terms}}$$

for all $i \in \mathbb{N}$.

Our goal is to show that $\forall n \geq 2 \in \mathbb{Z}$, there exists some m_n such that

$$x_{m_n} = x_{m_n+1} = \dots \pmod{n}.$$

We proceed with strong induction on n . The base case is $n = 2$, in this case, $x_1 = 2, x_2 = 2^2 = 4, x_3 = 2^{2^2} = 16, \dots$, so $m_1 = 1$, the statement holds for $n = 2$.

The inductive hypothesis: assume this works for all integers $2, \dots, n-1$ for some integer $n \geq 3$, we want to show that there is also a desired integer m_n .

We can split $n = 2^a b$ to divide and conquer using the Chinese Remainder Theorem, considering the sequence mod 2^a and the sequence mod b , and deducing the behavior under mod n .

Under mod 2^a , for all $j \in \mathbb{N}$ such that $x_{j-1} \geq a$, $x_j = 0 \pmod{2^a}$, this is because $x_j = 2^{x_{j-1}}$ and we need to ensure that $x_{j-1} \geq a$, so we set m_a to be the smallest j such j .

Under mod b , we note that $\gcd(b, 2) = 1$, so from Euler's theorem:

$$2^{\phi(b)} = 1 \pmod{b}$$

then for $x_j = x_{j+1} \pmod{b}$, we only need to ensure that $2^{x_{j-1}} = 2^{x_j} \pmod{b}$, so we need

$$x_{j-1} = x_j \pmod{\phi(b)}.$$

Here comes the punchline! Since $\phi(b) < b < n$, there exists $m_{\phi(b)}$ such that

$$x_{m_{\phi(b)}} = x_{m_{\phi(b)+1}} = \dots \pmod{\phi(b)},$$

then we take $m_b = m_{\phi(b)+1}$, so that

$$x_{m_b} = x_{m_b+1} = \dots \pmod{b}.$$

Now define $m_n = \max(m_a, m_b)$, then

$$x_{m_n} = x_{m_n+1} \pmod{2^a}$$

$$x_{m_n} = x_{m_n+1} \pmod{b}$$

hence $2^a \mid x_{m_n+1} - x_{m_n}$ and $b \mid x_{m_n+1} - x_{m_n}$, and since $\gcd(2^a, b) = 1$, $n = 2^a b \mid x_{m_n+1} - x_{m_n}$ hence

$$x_{m_n} = x_{m_n+1} = \dots \pmod{n}.$$

Furthermore, since m_b is always larger than m_a , $m_n = m_b \leq b < n$, so the original claim always holds.